

Sachin N. Redbook - Abstracts

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SYNOPSIS CHAMPIONSHIP PROJECT ABSTRACT

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With the instances of service disruptions from large-scale cloud systems such as IBM, user experience is negatively impacted, and consequences are increased without accurately and quickly identifying these anomalous instances. The integration of automated screening processes can aid in the accuracy and time taken to predict these anomalies and become aware of the issue at hand. Our study aims to provide a robust solution to anomaly detection in cloud infrastructure systems. We used a multistep optimization and filtering pipeline, to improve upon preliminary results from Islam et. al, who provided the dataset we used in our anomaly detection process and performed rudimentary analysis. For example, our process included a feature scoring algorithm, a two-step iterative neural network system, and a final anomaly instance prediction model based on density of nearby flagged indices. We were able to achieve significantly greater results because we managed to avoid weaknesses presented by a single neural network structure, presenting an argument for a more widespread use of such layered approaches. We successfully predicted 16/25 ground truth anomalies labeled in the dataset, improving the initial 6/25 anomalies predicted in the initial research; we also attained a greater ratio of true positives to false positives in our results. Cloud-based systems can be positively impacted by the incorporation of the kind of deep learning approach presented in this research, as it can help with preliminary anomaly detection so that operation teams can be informed of suspicious time windows.

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(Project A59)

Vihaan Parikh

ROCA: A Dynamic Metaheuristic Framework for Route Optimization Using Context-Aware Simulation of Ant Colony Systems

Travelling Salesman Problem (TSP) is a combinatorial optimization problem involving cyclic route planning, arising across a wide range of fields including emergency dispatch, networking, and computer-aided manufacturing. Due to superpolynomial time complexity, it is impractical to solve TSP via brute force. Over the last 40 years, many algorithms have been developed to find close-to-optimal solutions. However, the success of these heuristics is contingent on the assumption of a static solution space during computation. In real world applications of TSP, this assumption is often inadequate: traffic flow, latency, and available resources change continuously. This research presents a robust, environment-agnostic framework surrounding Ant Colony Optimization (ACO), a swarm-intelligence approach to TSP, designed to solve under dynamic costs and constraints. First, ACO's episodic structure cannot attribute specific choices to a "good" or "bad" solution. By introducing mid-cycle feedback, ROCA evaluates decisions at finer granularity. Second, metadata associated with each choice enables human language constraints to be injected directly into the algorithm's probabilistic selection step. Third, self-evolving parameters allow ROCA to converge more efficiently, independent of solution space uniformity. Compared to a contemporary ACO baseline implementation, ROCA reduced solution variability by an average of 24.58% and converged within 5% of newly introduced optima. Additionally, between 84.6-92.4% of contextual constraints were successfully incorporated. Evolutionary approaches outperformed their counterparts by roughly 4%. These improvements translate to faster emergency response times, saved compute, and accelerated production in time varying routing environments, laying groundwork for future domain-specific systems that fully bridge the theory-to-practice gap.

Real-time Disaster Search & Rescue System utilizing Semantic Segmentation and Object Recognition onboard Fixed-wing UAV

Rufan Yin

Stratford Preparatory Blackford, California, CA

Traditional search-and-rescue (SAR) operations rely heavily on manual aerial interpretation and often fail under low-visibility or communication-denied conditions. This project aims to address the engineering challenge of performing multimodal survivor detection under extreme pixel sparsity and strict edge power constraints, without reliance on cloud infrastructure.

A fully onboard artificial intelligence system was engineered on an NVIDIA Jetson Orin NX platform with a custom TensorRT-accelerated inference pipeline. The vision subsystem integrates a modified YOLOv11 detector optimized for sub-10-pixel aerial human targets and a custom Swin Transformer-based terrain segmentation model. Both networks were restructured using FP16 quantization, layer fusion, and pipeline parallelization to maximize performance per watt and minimize latency. Visual inference was integrated with a four-microphone array for voice direction-of-arrival estimation and a wireless network interface for RF presence detection through a custom weighted multimodal fusion architecture.

Engineering validation emphasized throughput, latency, and power efficiency under full multimodal load. The system achieved a 95th percentile throughput of 156 frames per second while maintaining a 95th percentile power draw of 25.0 W, satisfying real-time flight constraints. Integrated mission testing confirmed synchronized cross-modal detection and successful human identification at 200 feet under both visible and partially occluded conditions. This test achieved a detection rate 2.6 times greater than that of a single-modality (vision-only) test.

By demonstrating that transformer-based segmentation, real-time object detection, and acoustic/RF sensing can operate concurrently within strict edge power budgets, this work establishes a scalable systems-engineering framework for resilient disaster-response UAV platforms.

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E13 Abstract

Project Title: A Convex Directional Spline Approach with Huber-Regularized Loss for Stable Deep Optimization

Vikram Atmuri, Grade 11, Santa Clara High School

Machine learning models power medical diagnosis, autonomous vehicles, and climate prediction, yet their accuracy is limited by static loss functions. Even a 1% accuracy gain could mean thousands of additional cancers detected early or millions saved in accident costs. Loss functions are formulas which guide a model to improve its predictions, and even the most widely used loss functions like Mean Squared Error (MSE) and Huber offer minimal mechanisms to adjust when data distributions shift.

This study proposes and evaluates a learnable Differing-Degree Spline loss function with Lagrangian dual variables that reshapes itself during training by learning optimal knot placements under monotonicity constraints, enabling it to adapt to data shifts in real-time. While prior work supports learnable loss parameters, this synthesis of Differing-Degree Splines with Lagrangian constraints for adaptive loss learning is novel.

The Lagrangian Differing-Degree Spline (LDDS) was evaluated under varying outlier contamination. Results showed average test improvements of 5-9% over MSE, and 4-7% over Huber across protein folding, diabetes, and image classification datasets. Evaluated on datasets with 40% outliers, LDDS achieved a 13.9% improvement while converging 11% faster than MSE. Further, dual variables enforced monotonicity, with violations converging to zero ($P = 0.012$ for MSE improvement; $P = 0.0011$ for constraint satisfaction).

These results suggest LDDS could result in higher accuracy under data distribution shifts. Moreover, it could cut training time by ~14% resulting in accelerated deployments of machine learning systems, reducing ~0.9 million metric tonnes of CO₂ caused by excess compute annually.

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L92

Optimizing Barter Trading Networks

Brandon Labio

Bartering, the oldest form of trade, still thrives today. An estimated \$14B worth of goods is exchanged annually in barter networks. This is despite the networks not solving the age-old Barter Trading Matching Problem (BTMP): find a trader that has what I want and wants what I have.

I developed five algorithms that find the single trader (denoted as BTMP1) or the set of traders (BTMPSet) that can exchange the maximum amount of items given a trader T.

The algorithms solve BTMP in different ways. One uses network flow, two use inverted-indices, one uses databases, and the last is brute-force.

I tested the algorithms on 16 datasets modeled after today's networks (e.g., Craigslist, Rehash). Our inverted-index based BTMP1 algorithm returns optimal trade proposals within 2ms on average that exchange 8% of a trader's inventory. For BTMPSet, our network-flow based algorithm returns optimal trade proposals within 13ms that exchange 83% of a trader's inventory. The heuristic inverted-index based BTMPSet algorithm returns proposals that exchange 73% of inventory but using 35% less traders.

I show through complexity analysis that the network flow and the inverted-index based algorithms scale with the ratio N/M (instead of N or M), where N is the number of traders, and M is the number of items in the network. And through experimentation, I show that these algorithms can handle sizes beyond the sizes of today's networks.

Using network flow and inverted-indices, I created scalable algorithms that make finding possible traders in barter networks much easier.

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This project is the first systematic attempt to study the distance distribution of Reed-Muller codes across all parameters, including large finite fields. Its practical applications are wide-ranging, most importantly in cryptography (identity systems, blockchain scalability) wherein distance distribution of codes helps determine the soundness of zero-knowledge proofs, in quantum computing to determine quantum code's error-correction capability, and in storage system design to extend flash memory lifetime. These innovations strive to make our computing infrastructure more secure, reliable, and efficient. In this project, I answer a fundamental and long-standing open question in coding theory called the distance distribution problem (or coset weight distribution problem), which asks how distances between words and codewords are distributed across the space of all possible messages. Although special cases such as weight distribution of Reed-Muller codes or distance distribution of Reed-Solomon codes have been studied in this regard, the general problem for Reed-Muller codes over arbitrary finite fields stands unsolved due to the complex nature of multivariate polynomials. I resolve this nearly 50-year-old open problem by giving the first general error bounds on the distance distribution of Reed-Muller codes. This is algebraically equivalent to counting multivariate polynomials over finite fields with prescribed degree, coefficients, and number of zeroes. The unique approach I take analyzes character sums over multivariate polynomial quotient rings over a finite field, which have not been studied in more than one variable prior to this work. As a corollary of this work, I derived error bounds for the asymptotics of the distance distribution.